



Master Thesis

Thesis Title: Concise and Engaging Title

by

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LLM Evaluators: teststs

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Abstract. The abstract should briefly summarize the contents of the paper in 150–250 words.

Keywords: First keyword · Second keyword · Another keyword.

1 Introduction

1.1 General Introduction

DRAFT

LLMs are increasingly used for content generation. However, reliability remains a bottleneck.

Subjective evaluation poses a challenge for Large Language Models (LLMs) and humans alike. Consider an evaluation task, such as the correctness of a mathematical derivation, the number of grammatical errors in a sentence, or the helpfulness of an LLM output. An evaluation task is subjective when the ground truth is not explicitly known, such as for the helpfulness of LLM output: what is helpful to one user might be irrelevant to another. Contrast this with objective evaluation, where output can be validated against an observable ground truth, as is generally the case for e.g. mathematical statements and grammar.

General practice in subjective evaluation is to ask humans to provide 'gold labels'. By training humans to follow a customized evaluation protocol, human evaluators are assumed to approximate a valid measurement of the latent ground truth. However, this process is unscalable.

To automate evaluation, classic Natural Language Generation (NLG) evaluation techniques such as BLEU and ROUGE were often used. Yet, because these only consider surface-level linguistic characteristics like n-gram overlap, they often suffer from low correlation with human evaluations [6]. While useful for simple tasks like translation and summarization quality, they fail to capture the linguistic nuances needed for more complex tasks.

LLMs offer a promising solution. In a seminal paper, Zheng et al. [13] use well-aligned LLMs such as GPT4 in pairwise comparisons, where two candidate texts are compared and the best one is selected. GPT-4 achieves over 80% agreement with human preferences, on par with human-human agreement. Liu et al. [6] adopt a similar approach in their G-Eval framework, prompting LLMs to grade text on a fine-grained Likert scale. Their approach shows significantly higher correlation with human judgements compared to previous state-of-the-art evaluators.

However promising, (limitations of current methods, bridge to subsection 1.2 industrial partner operational challenges)

1.2 Industrial Partner: Barter

The current study is motivated by the operational challenges faced by the industrial partner, Barter. The company aims to leverage Generative AI (GenAI) to streamline the onboarding experience, specifically addressing the friction new partners experience during the sign-up flow. Currently, users must manually write marketing copy such as company descriptions and deal offers before they can make use of the platform. This creates a significant barrier to entry, resulting in user attrition.

While GenAI offers a scalable solution to automate this drafting process, its implementation in business contexts is hindered by the inherent variability of output quality. As LLMs are probabilistic in nature, some degree of inconsistency is unavoidable. Although well-engineered prompts might yield high-quality output in the majority of cases, the accumulation of low-quality output (such as hallucinations and misalignment) becomes a significant operational problem at high volumes.

Two fundamental challenges lie at the core of this problem. First is the challenge of identification: without a reliable mechanism to flag low-quality output, we are unable to quantify the severity of the problem, or make the problematic output actionable. The second is the challenge of diagnosis: improving the output requires an evaluation to identify the reasons why the quality is insufficient. For GenAI to function as an automated solution, it must not only detect the errors but also provide the feedback to resolve them.

Therefore, although Barter's problem statement concerns generation, the solution to the problem begins with accurate validation. Currently, Barter operates in a similar fashion, where employees use GenAI to generate content which is manually evaluated and refined. However, such a human-in-the-loop (HITL) approach is costly and unscalable. To address this, this thesis explores whether small-scale open-source LLMs can be used for automated "AI-in-the-loop" (AITL) evaluation. We introduce a method inspired by the latent variable models of Psychometrics to provide the granular evaluation required for the identification and diagnosis of output quality.

2 Literature Review

2.1 Traditional NLP evaluation methods

2.2 LLM-as-a-Judge

2.3 Measurements through Analytic Decomposition

Prior work provides evidence that analytic decomposition can improve rater reliability or agreement on the decomposed dimensions, but it rarely tests whether aggregating decomposed judgments yields a superior measurement of the corresponding holistic construct compared to a direct holistic evaluation.

3 Theoretical Framework

- NOTE: In the final version, some of the information below might be moved to Introduction/Literature Review

This thesis treats LLM-based evaluators as measurement instruments that map an input (e.g., marketing copy text) to a score that represents an underlying (subjective) construct (e.g., "Quality"). The goal is to align the LLM evaluation with human evaluation.

The construct of interest can be evaluated directly, in which case it is considered holistically. However, a fundamental challenge in evaluating subjective attributes is that it is not a singular, physical property like temperature or length, but a high-dimensional construct. We propose a methodology that decomposes the construct into constituent sub-dimensions. The core hypothesis is that by modelling the target variable as a composite of these constituents, we obtain a better measurement of the whole.

A similar measurement approach is taken in Li and Sun [4] and Wang et al. [12]. Li and Sun [4] note that it is difficult to align general attributes such as "usefulness" and "harmlessness" with human evaluations, so they decompose them into more specific criteria. Wang et al. [12] find that an "analytic scoring" approach outperforms holistic scoring, where they supply the LLM with Rubrics containing multi-dimensional assessment criteria on which it bases its evaluation.

However, these studies typically treat decomposition as a pragmatic scoring or training design choice, rather than as an explicit construct-specification problem. As a result, it remains unclear how the decomposed scores should be interpreted, and which criteria for validity follow. Validity in psychometrics concerns whether an instrument measures what it intends to measure. By making the measurement model explicit, we can define the requirements for meaningful measurements and derive a validation strategy that matches the construct specification.

Moreover, concerning operationalization, a common engineering strategy in LLM-based evaluation introduces construct-irrelevant variance due to two methodological choices. Firstly, they supply multiple evaluation dimensions in a single prompt. Positional bias is a common problem in LLMs, where the output is dependent on the order in which the information is presented (see Gu et al. [3]). Secondly, they require the LLM to output all evaluations within a single generation sequence. This introduces autoregressive dependencies, so earlier generated scores condition the token distribution over subsequent scores. To mitigate these issues we measure each dimension in isolation, which is consistent with Bang et al. [1], who also measure criterion-specific scores via separate prompts. Our contribution is to formalize why this operationalization matters: independent prompting mitigates (1) positional effects (2) autoregressive coupling, while additionally enabling (3) logit-level analysis and (4) reducing per-inference task complexity, aligning with the constraints of small open-source models in our production environment.

Finally, the validation methodologies in Li and Sun [4] and Wang et al. [12] require improvement. Firstly, they establish *convergent validity* by comparing LLM ratings to human ratings on identical analytic scales. This approach demonstrates that an LLM can mimic human grading on specific sub-dimensions, but fails to prove that the decomposition strategy itself yields a superior measurement of the overall quality compared to a holistic evaluation. The theoretical foundation from Section ?? allows us to specify a model to directly assess this. Furthermore, neither paper has access to a dataset which includes behavioral outcomes, limiting their validation to human proxy labels. Demonstrating functional utility is especially important considering the intended use of the evaluator on the Barter platform. We extend validation by assessing *predictive validity* against a downstream behavioral KPI.

Accordingly, the structure of this chapter is as follows.

- **Theoretical definition:** We explain the two causal models that define the relationship between a composite construct ("latent variable") and its constituents ("indicators"): the *reflective* and the *formative* model.
- **Computational operationalization:** We formalize the independent measurement of these constituents by defining the mathematical mapping of dimensions to logit-based scores.
- **Empirical validation:** We explain how we assess measurement validity through the psychometric concept of *construct validity*: whether an instrument truly measures the phenomenon it is supposed to measure. We assess construct validity through *convergent validity* (alignment with human holistic labels) and *predictive validity* (prediction of a behavioral KPI).

Theoretical definition: Latent Variable Models Latent variable models define a relationship between a target construct and its indicators. Two model types are distinguished: formative and reflective [2]. They differ in their assumptions about the direction of causality.

In formative models, the latent variable is assumed to be a composite construct, caused by the indicators ("causal indicators"). The canonical example is Socioeconomic Status (SES), which can be defined as the composite of three main features: income, education, and prestige [7]. Formally, for a set of causal indicators indexed by $i = 1, 2, \dots, n$, we assume

$$\eta = \sum_{i=1}^n \gamma_i x_i + \zeta \quad (1)$$

where η is a latent construct, γ_i a formative weight, x_i a manifest variable, ζ the disturbance term.

In reflective models, the latent variable causes the manifest variables ("effect indicators"). For example, following a "g-factor" interpretation of the latent variable intelligence [9], we assume that intelligence is an inherent trait that humans possess which causes people's performance on cognitive tasks. Formally, for any given manifest variable i , we assume

$$x_i = \lambda_i \eta + \varepsilon_i \quad (2)$$

where x_i is an effect indicator, λ_i a factor loading, η the latent construct, and ε_i the measurement error.

Exemplars in LLM-based measurement. Decomposition already appears in LLM-based evaluation, but with different implicit measurement stances. Wang et al. [12] study LLM scoring in large-scale writing assessment using psychometric reliability frameworks, consistent with treating rubric dimensions as manifestations of a common underlying writing ability (reflective stance). Li and Sun [4] operationalize broad evaluation targets through batteries of criteria organized by task category, emphasizing domain coverage, which is consistent with treating evaluation as a composite defined by its constituents (formative stance). These two uses illustrate why decomposition should be paired with an explicit construct specification: it determines both interpretability and what constitutes appropriate validity evidence.

Implications. These contrasting stances illustrate that the assumed causal model has downstream consequences for both measurement design and validation. Under a reflective specification, indicators are treated as effects of a common underlying attribute; accordingly, strengthening the measurement design (e.g., additional tasks/raters, well-functioning indicators) improves the precision of the latent estimate without redefining the construct, and internal-structure evidence (e.g., factor structure and reliability/generalizability) is a primary diagnostic of measurement quality. Under a formative specification, the construct is defined by its constituents; indicators are not expected to be interchangeable or strongly intercorrelated, and adding or omitting an indicator changes the construct’s meaning. Measurement quality therefore hinges on (i) defensible construct specification and content coverage and (ii) accurate measurement of each constituent, while validation relies primarily on external relations (e.g., alignment with alternative measures of the target attribute and prediction of relevant criteria).

Construct specification (quality-for-purpose). In this thesis, “marketing copy quality” is treated as a quality-for-purpose construct: the attribute is defined by the requirements of its intended use in the Barter setting (i.e., what stakeholders need the copy to achieve). Consequently, the construct is not assumed to be a single underlying trait that causally manifests in multiple observed ratings, but a composite defined by a set of constituent criteria (e.g., clarity, persuasiveness, and compliance with offer details) selected to cover the relevant content domain. This motivates a formative specification: the indicator set constitutes the construct definition, so modifying the set changes what “quality” means. The role of empirical analysis is therefore not to discover a latent factor behind the indicators, but to test whether the specified composite yields valid measurements with respect to human judgments and downstream behavioral outcomes.

3.1 Operationalization: LLM as a Measurement Instrument

Following the formative specification defined in Section ??, we must define a computational process that measures each indicator x_i (corresponding to criterion c_i) independently. We operationalize the LLM not merely as a generative agent, but as a probabilistic measurement instrument. This section formalizes the mapping from input text to score and demonstrates why standard generation techniques introduce construct-irrelevant variance (CIV) through positional and autoregressive dependencies.

Input Specification (Encoding) Formally, an LLM maps an input token sequence $\mathbf{u}$ to a distribution of next-token logits $\mathbf{z}(\mathbf{u}) \in \mathbb{R}^{|\mathcal{V}|}$ over a vocabulary $\mathcal{V}$. The conditional probability of a token v is given by the softmax function:

$$P(v \mid \mathbf{u}) = \frac{\exp(z_v(\mathbf{u}))}{\sum_{w \in \mathcal{V}} \exp(z_w(\mathbf{u}))}. \quad (3)$$

In standard evaluation pipelines, multiple criteria are often concatenated into a single input sequence $\mathbf{U}(d) = [s; c_1; \dots; c_J; d]$. However, due to the self-attention mechanism, the representation of any specific criterion c_j becomes a function of its position within $\mathbf{U}$. This introduces positional bias, where the measurement is sensitive to the order of instructions rather than the content of the document d [3].

To ensure the measurement of any indicator is invariant to the presence or order of other indicators, we define the input strictly as the triplet of task instruction s , single criterion c_j , and document d . For each criterion j , we construct an independent input sequence $\mathbf{u}_j$:

$$\mathbf{u}_j(d) = [s; c_j; d]. \quad (4)$$

This ensures that the logit vector $\mathbf{z}(\mathbf{u}_j)$ depends exclusively on the definition of c_j and the document content, isolating the indicator from contextual noise.

Score Extraction (Decoding) Standard approaches often require the model to generate a sequence of scores $\mathbf{y} = (y_1, \dots, y_J)$ for the batched criteria. This relies on autoregressive decoding, where the probability of a score y_t is conditioned on the previously generated scores $y_{<t}$:

$$P(\mathbf{y} \mid \mathbf{U}) = \prod_{t=1}^J P(y_t \mid \mathbf{U}, y_{<t}). \quad (5)$$

This formulation introduces two sources of construct-irrelevant variance. First, it creates *autoregressive coupling*: the measurement of the t -th indicator is mathematically dependent on the values assigned to previous indicators, violating the requirement that a measurement should depend only on the attribute being measured. Second, the generation of a multi-token sequence relies on stochastic sampling strategies (e.g., nucleus sampling) and decoding hyperparameters (e.g., temperature), introducing stochasticity that degrades measurement consistency.

To eliminate these dependencies, we restrict the generation horizon to a single token. We treat the evaluation of criterion c_j as an independent classification task and extract the score y_j by directly analyzing the next-token distribution conditioned on $\mathbf{u}_j(d)$:

$$y_j \sim P(\cdot \mid \mathbf{u}_j(d)). \quad (6)$$

By constraining the "generation" to the first step, we ensure that $P(y_j)$ is conditioned strictly on the input $\mathbf{u}_j(d)$ and never on prior outputs. Furthermore, this approach grants direct access to the logits $\mathbf{z}(\mathbf{u}_j)$, allowing us to bypass the sampling step entirely and derive deterministic, reproducible measurements invariant to decoding strategies.

3.2 Empirical Validation

Having operationalized the measurement instrument, we must now assess its validity. In psychometrics, *validity* refers to the degree to which evidence supports the interpretation of test scores for their proposed uses [8]. Since "Marketing Copy Quality" is a latent variable with no single physical ground truth, we cannot validate it directly. Instead, we adopt a *construct validity* approach, seeking evidence that our Formative Model behaves as a valid measure of quality should.

We assess validity through two complementary frameworks: *Convergent Validity* (alignment with human expert judgment) and *Predictive Validity* (alignment with downstream behavioral outcomes).

Convergent Validity: Alignment with Human Judgment Convergent validity assesses whether a new instrument correlates with established measures of the same construct. While human evaluation is subjective and resource-intensive, it remains the standard proxy for "true" quality in Natural Language Generation tasks.

We compare two measurement models against human holistic labels:

1. **The Holistic LLM Model:** A standard approach where the LLM is prompted to provide a single quality score directly.
2. **The Formative LLM Model (Ours):** The composite score derived from the weighted aggregation of the independent indicator measurements defined in Section 3.1.

Our hypothesis is that the Formative Model, by explicitly decomposing the construct into sub-dimensions (e.g., fluency, persuasiveness), will yield a higher correlation with human judgments than the Holistic Model. This effectively tests whether the "deconstruction" strategy aids the LLM in mirroring human evaluative logic.

Predictive Validity: Functional Utility Since we define quality as "quality-for-purpose" (Section 3), the ultimate test of validity is functional utility: does the measured quality predict the actual effectiveness of the marketing copy?

This constitutes *predictive validity*. We utilize a dataset of marketing emails with associated behavioral KPIs (e.g., Click-Through Rate or Conversion Rate). We assess the ability of the Formative Model to explain variance in these behavioral outcomes compared to the Holistic Model. Strong predictive validity would demonstrate that our decomposition strategy captures the specific attributes that drive real-world performance, arguably the most critical requirement for an automated evaluator in a production environment.

4 Methods

4.1 Data

FeedbackQA The *FeedbackQA* dataset Li et al. [5] contains evaluations of Q&A pairs ($n = 5660$). Q&A pairs related to Covid-19 were scraped from the websites of WHO, US Government, UK Government, Canadian government, and the Australian government. Crowd-source workers from English-speaking countries were hired through Amazon’s MTurk platform to rate each Q&A pair and provide feedback. The rating is performed on a 4-point likert scale (*bad, could be improved, good, excellent*). The feedback concerns "a natural language explanation elaborating on the strengths and/or weaknesses of the answer" [5]. Since each Q&A pair contains two ratings, we aggregate both into the mean to obtain our "ground truth" target variable.

Barter deals The *Barter Deals* dataset contains 'deals' written by companies ($n = ???$). On the Barter platform, companies ('partners') can barter goods and services in exchange for influencer content. For example, a company that sells hair styling tools offers their new product in exchange for a video of a creator using it; or, a museum provides free entrance to a creator and their friends in exchange for a video of their visit. The 'deal text' is the marketing copy that is posted on the Barter platform. Users can apply to deals, and the partner can accept or reject the application. Once accepted, the partner provides the good or service, and the creator provides the content.

In the current study, we consider the 'deal text' (marketing copy) and the number of applications the deal has obtained after the first seven days of going live (*applications after 7 days*). The variable *applications after 7 days* is chosen because it serves as a behavioral proxy for marketing copy quality. One might apply to a deal because they like the product or the brand, but they could also be convinced to apply because of the way the copy is written. A valid measurement of marketing copy quality would therefore be predictive of whether somebody applies to a deal or not. By controlling for the fixed effects of the brand and the product category we can separate the effects of marketing copy quality, and hereby determine whether the LLM measurements satisfy predictive validity.

4.2 Operationalization of Evaluation Constructs

To empirically measure the efficacy of the formative prompting framework, we define an experimental condition and a control condition (baseline). In the experimental condition (*formative prompt*) we treat the construct as a formative variable by decomposing it into sub-dimensions (e.g., d_1, d_2, d_3). We prompt the LLM to measure the subdimensions individually, yielding multiple scalar values. In the control condition (*holistic prompt*) we treat the construct as a monolith, forcing the model to implicitly aggregate the dimensions into a single scalar rating Y .

Construct definition: FeedbackQA For the question answering task, we derive the formative dimensions from the qualitative analysis of the feedback provided in Li et al. [5]. The authors identify numerous criteria that define "answer quality" by examining the explanations crowd-sourced workers provided for their ratings. From this, we derive three dimensions:

- **Relevance:** The extent to which the answer avoids irrelevant content
- **Completeness:** Whether the answer covers all aspects of the query.
- **Directness:** Whether the answer directly answers the question.

We expect a degree of covariance among the dimensions, since high-quality answers generally score high across all three. However, the formative framework allows for orthogonal variation across the dimensions as well. For example, Li et al. [5] highlight an answer rated *could be improved* (2/4) with the feedback: *"The answer is about some of the online risks but not about how to protect against them."* This indicates an answer that is relevant but incomplete. While a holistic prompt forces the model to compress the conflicting signals into a single scalar value, the formative prompt explicitly models this by decomposing it.

Construct definition: Barter deals DRAFT:

In the absence of prior academic literature for the proprietary Barter dataset, we derived the dimensions through stakeholder analysis and domain requirements. The "Marketing Effectiveness" construct is decomposed into:

- **Persuasiveness:** The strength of the call-to-action.
- **Tone Alignment:** The consistency with the brand’s requested voice.
- **Clarity:** The readability of the deal proposition.

4.3 Model selection

To perform evaluations we select three popular small-scale, open-source instruction-tuned models: Qwen3-4B-Instruct-2507 [11] (AliBaba), Gemma3-4B-IT [10] (Google), and Llama-3.2-3B-Instruct (Meta). These models were selected to represent the current state-of-the-art open-source models.

The choice of model type (small, open-source, instruction-tuned) is based on the methodological requirements and the operational constraints of the industrial partner. Methodologically, the use of open-source models enables "white-box" access to the model’s probability distribution. This allows us to compute the logit-weighted mean and the Shannon entropy. Moreover, we prefer instruction-tuned models over base models to ensure adherence to the evaluation instructions provided in the prompt. Operationally, Barter requires the evaluator to balance validity with cost-efficiency and latency. Proprietary models like GPT-4 incur costs that scale with volume, while open-source models incur only fixed compute costs, making them a more scalable solution.

4.4 Experimental design & Validation Strategy

Convergent Validity: Alignment with Human Raters (FeedbackQA)

Predictive Validity: Behavioral Prediction (Barter Deals)

5 Results

5.1 Academic validation (FeedbackQA)

5.2 Industrial application (Barter)

6 Discussion

"We assumed robustness; future work should stress-test it."

"A limitation of this study is the absence of adversarial robustness testing. While the method demonstrates high concurrent validity on the FeedbackQA dataset, we did not evaluate the model's stability against adversarial perturbations (e.g., randomized word order or irrelevant insertions). Future work should investigate whether the formative latent variables remain stable under such conditions to rule out spurious correlations."

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